EXPERIMENT TO UNDERSTAND KNN ALGORITHMN FOR A GIVEN DATA SET

PROGRAM:

```
import numpy as np
import pandas as pd

df = pd.read_csv('Iris_modified.csv')
df
df.info()
df.variety.value_counts()
df.head()
features = df.iloc[:, :-1].values
label = df.iloc[:, 4].values
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier

xtrain, xtest, ytrain, ytest = train_test_split(
    features, label, test_size=0.2, random_state=42
)
print(model_KNN.score(xtrain, ytrain))
print(model_KNN.score(xtest, ytest))
from sklearn.metrics import confusion_matrix

confusion_matrix(label, model_KNN.predict(features))
from sklearn.metrics import classification_report
```

print(classification_report(label, model_KNN.predict(features)))

OUTPUT:

	sepal.length	sepal.width	petal.length	petal.width	variety
0	5.2	3.5	1.5	0.3	Setosa
1	4.7	3.2	1.5	0.3	Setosa
2	4.6	3.2	1.2	0.4	Setosa
3	4.7	3.2	1.5	0.1	Setosa
4	5.0	3.4	1.5	0.3	Setosa
...	...	...	...	...	...
145	6.7	3.0	5.0	2.3	Virginica
146	6.5	2.6	4.9	1.8	Virginica
147	6.4	2.8	5.4	2.0	Virginica
148	6.1	3.4	5.2	2.2	Virginica
149	6.0	3.1	5.3	1.8	Virginica

150 rows × 5 columns

```
<class 'pandas.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   sepal.length  150 non-null    float64
 1   sepal.width   150 non-null    float64
 2   petal.length  150 non-null    float64
 3   petal.width   150 non-null    float64
 4   variety       150 non-null    str
dtypes: float64(4), str(1)
memory usage: 6.0 KB
```

```
variety
Setosa        50
Versicolor    50
Virginica     50
Name: count, dtype: int64
```

	sepal.length	sepal.width	petal.length	petal.width	variety
0	5.2	3.5	1.5	0.3	Setosa
1	4.7	3.2	1.5	0.3	Setosa
2	4.6	3.2	1.2	0.4	Setosa
3	4.7	3.2	1.5	0.1	Setosa
4	5.0	3.4	1.5	0.3	Setosa

```
0.9416666666666667
0.9666666666666667
```

```
array([[50,  0,  0],
       [ 0, 48,  2],
       [ 0,  6, 44]])
```

```
  precision    recall  f1-score   support

      Setosa       1.00      1.00      1.00        50
  Versicolor       0.89      0.96      0.92        50
   Virginica       0.96      0.88      0.92        50

    accuracy                           0.95       150
   macro avg       0.95      0.95      0.95       150
weighted avg       0.95      0.95      0.95       150
```